# Escape speed

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain escape in an ideal gravitational-energy model.

**SUCCESS CRITERIA - I can...**

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A 3 kg cart at 10 m/s hits an identical cart at rest. Final velocities are 4 and 6 m/s. Check momentum and energy; classify the collision.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

**Worked reasoning:** For a toy planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

**Your turn:** Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Does escaping mean that gravity turns off at the launch point? Explain what the speed allows in an energy model.

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

For $GM=400\ m^3/s^2$ and r=8 m, calculate circular speed and escape speed. Compare their ratio.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

### 05 Analyze the reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 6 kg object descends 8 m near Earth. Use g=10. Find ΔU and the work done by gravity.

**Marzano check - circle:** 1 2 3

### 07 Exit check

At fixed GM, launch distance increases by factor 4. Compare escape speeds.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A planet's mass doubles while radius stays fixed. Compare escape speed. Then double both mass and radius and compare again.

**Marzano check - circle:** 1 2 3